III. Entanglement

- if a pure state $|\psi\rangle_{AB}$ on systems A, B cannot be written as $|\varphi\rangle_A \otimes |\Phi\rangle_B$, it is entangled.

The Bell States are the special ones.

There are four so called Bell-States that are maximally entangled and build an orthonormal basis:

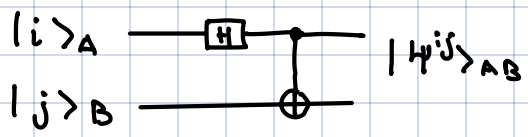
$$|\psi^{00}\rangle := \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \qquad |\psi^{01}\rangle := \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

$$|\psi^{01}\rangle := \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) \qquad |\psi^{11}\rangle := \frac{1}{\sqrt{2}}(|01\rangle - |10\rangle)$$

in general we can write: $|\psi^{ij}\rangle = (I \otimes \sigma_x^j \sigma_z^i)|\psi^{00}\rangle$

$$|\psi^{00}\rangle = |++\rangle + |--\rangle$$

Creation of Bell States



initial state $\lvert i,j\rangle_{AB}$		$(H_A \otimes I_B)\lvert i,j\rangle_{AB}$
$\lvert 00\rangle$		$(\lvert 00\rangle + \lvert 10\rangle)/\sqrt{2}$
$\lvert 01\rangle$	$\xrightarrow{H_A}$	$(\lvert 01\rangle + \lvert 11\rangle)/\sqrt{2}$
$\lvert 10\rangle$		$(\lvert 00\rangle - \lvert 10\rangle)/\sqrt{2}$
$\lvert 11\rangle$		$(\lvert 01\rangle - \lvert 11\rangle)/\sqrt{2}$

continued bellow...

She is doing it very fast, I have trouble following it, let's check what's happening

$$|00\rangle = |0\rangle \otimes |0\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}1\\0\\0\\0\end{pmatrix}$$

$$H_A \otimes I_B = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1\\ 1 & -1\end{pmatrix} \otimes \begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1 & 0 & 0\\ 1 & -1 & 0 & 0\\ 0 & 0 & 1 & 1\\ 0 & 0 & 1 & -1\end{pmatrix}$$

$$\Rightarrow H_A \otimes I_B\, |00\rangle = \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix}1 & 1 & 0 & 0\\ 1 & -1 & 0 & 0\\ 0 & 0 & 1 & 1\\ 0 & 0 & 1 & -1\end{pmatrix}}_{4\times 4}\underbrace{\begin{pmatrix}1\\0\\0\\0\end{pmatrix}}_{4\times 1} = \underbrace{\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\\0\\0\end{pmatrix}}_{4\times 1} = \frac{1}{\sqrt{2}}|00\rangle + |10\rangle$$

Consider that

$$|00\rangle = \begin{pmatrix}1\\0\\0\\0\end{pmatrix} \text{ kinda } 1$$

$$|01\rangle = \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\0\\1\\0\end{pmatrix} \text{ kinda } 3$$

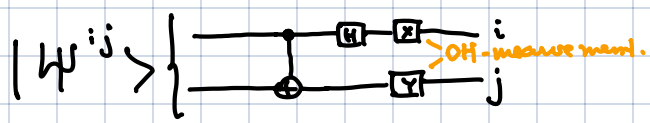
$$|10\rangle = \begin{pmatrix}0\\1\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} = \begin{pmatrix}0\\1\\0\\0\end{pmatrix} \text{ kinda } 2$$

$$|11\rangle = \begin{pmatrix}0\\1\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix} = \begin{pmatrix}0\\0\\0\\1\end{pmatrix} \text{ kinda } 4.$$

continued from above...

initial state $\lvert i,j\rangle_{AB}$	$(H_A \otimes I_B)\lvert i,j\rangle_{AB}$		
$\lvert 00\rangle$	$(\lvert 00\rangle + \lvert 10\rangle)/\sqrt{2}$	$\xrightarrow{CNOT_{A,B}}$	$(\lvert 00\rangle + \lvert 11\rangle)/\sqrt{2} = \lvert\Psi^{00}\rangle$
$\lvert 01\rangle$ $\xrightarrow{H_A}$	$(\lvert 01\rangle + \lvert 11\rangle)/\sqrt{2}$		$(\lvert 01\rangle + \lvert 10\rangle)/\sqrt{2} = \lvert\Psi^{01}\rangle$
$\lvert 10\rangle$	$(\lvert 00\rangle - \lvert 10\rangle)/\sqrt{2}$		$(\lvert 00\rangle - \lvert 11\rangle)/\sqrt{2} = \lvert\Psi^{10}\rangle$
$\lvert 11\rangle$	$(\lvert 01\rangle - \lvert 11\rangle)/\sqrt{2}$		$(\lvert 01\rangle - \lvert 10\rangle)/\sqrt{2} = \lvert\Psi^{11}\rangle.$

↳ opposite direction: Bell measurement.



⇒ these are classical outcomes. Correspond to a measurement of the state $|\Psi^{ij}\rangle$

Teleportation

- Goal: Alice wants to send her (unknown state)

$$|\varphi\rangle_S := \alpha|0\rangle_S + \beta|1\rangle_S.$$

to Bob. She can only send him two classical bits.

They both share the maximally entangled state

$$|\Psi^{00}\rangle_{AB} = \frac{1}{\sqrt{2}}\left(|00\rangle_{AB} + |11\rangle_{AB}\right).$$

$\Rightarrow$ initial state of the total system : $|\varphi\rangle_S \otimes |\Psi^{00}\rangle_{AB} = \frac{1}{\sqrt{2}}\big(\alpha|000\rangle_{SAB} \oplus$

$\oplus\, \alpha|011\rangle_{SAB} \oplus \beta|100\rangle_{SAB} + \beta|111\rangle_{SAB}\big) =$

$$= \frac{1}{2\sqrt{2}}\Big[\left(|00\rangle_{SA} + |11\rangle_{SA}\right)\otimes\left(\alpha|0\rangle_B + \beta|1\rangle_B\right) + \left(|01\rangle_{SA} + |10\rangle_{SA}\right)\otimes\left(\alpha|1\rangle_B + \beta|0\rangle_B\right)$$

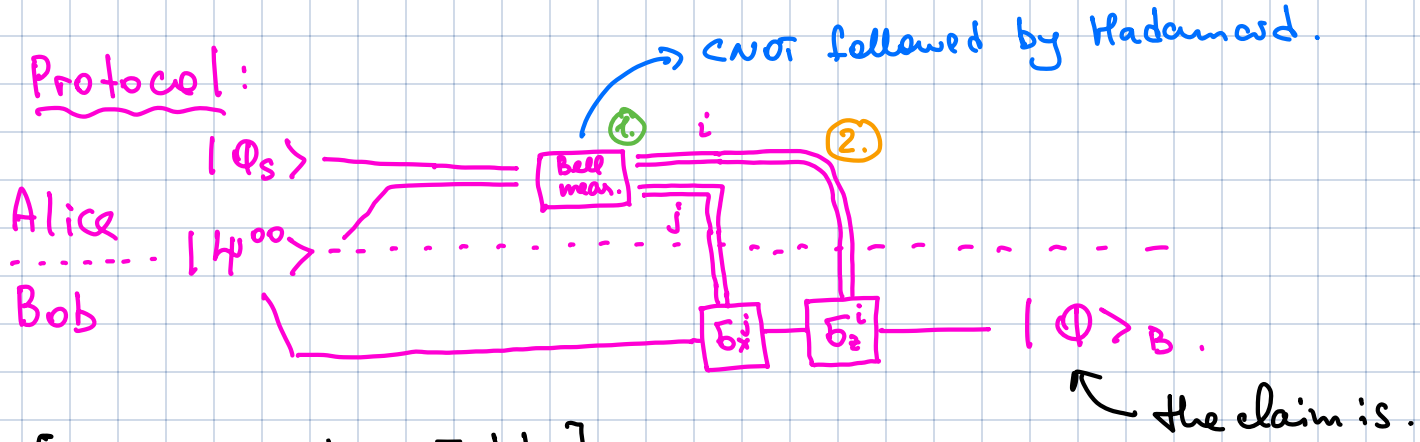
$$+ \left(|00\rangle_{SA} - |11\rangle_{SA}\right)\otimes\left(\alpha|0\rangle_B - \beta|1\rangle_B\right) + \left(|01\rangle_{SA} - |10\rangle_{SA}\right)\otimes$$

$$\otimes\left(\alpha|1\rangle_B - \beta|0\rangle_B\right)\Big] =$$

$$= \frac{1}{2}\Big[|\Psi^{00}\rangle_{SA}\otimes|\varphi\rangle_B + |\Psi^{01}\rangle_{SA}\otimes\sigma_x|\varphi\rangle_B + |\Psi^{10}\rangle_{SA}\otimes\sigma_z|\varphi\rangle_B +$$

$$+ |\Psi^{11}\rangle_{SA}\otimes\sigma_x\sigma_z|\varphi\rangle_B\Big]$$

Protocol:



[Let's consider a Table]

1. Alice measures on S&A in Bell basis.
2. She sends her classical outputs i,j to Bob.
3. Bob applies $\sigma_z^i \cdot \sigma_x^j$ to his qubit and sets $|\varphi\rangle$!

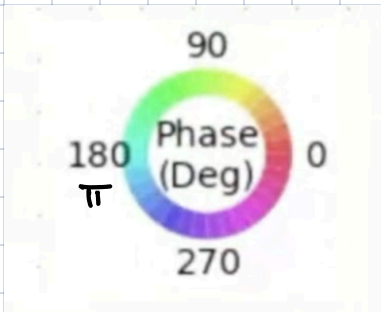
1. Alice's meas.	$\rightarrow$ Bob's state.	2. Alice sends i,j	3. Bob applies	$\rightarrow$ Bob's final stage
$\|\Psi^{00}\rangle$	$\|\varphi\rangle_B$	0 0	I	$\|\varphi\rangle_B$
$\|\Psi^{01}\rangle$	$\sigma_x\|\varphi\rangle_B$	0 1	σ_x	$\|\varphi\rangle_B$
$\|\Psi^{10}\rangle$	$\sigma_z\|\varphi\rangle_B$	1 0	σ_z	$\|\varphi\rangle_B$
$\|\Psi^{11}\rangle$	$\sigma_x\sigma_z\|\varphi\rangle_B$	1 1	$\sigma_z\sigma_x$	$\|\varphi\rangle_B$

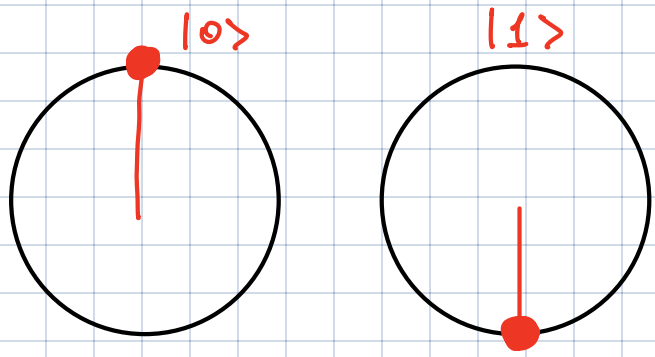
Note that Alices state collapsed during the measurement.
She does not have the initial state anymore.
This is expected due to the no-cloning theory.
As she cannot copy her state, but just send her state to Bob when destroying her own.

Q-Sphere

- Bloch sphere can only illustrate the state of 1 qubit.
  => for multiple qubits: Q-sphere

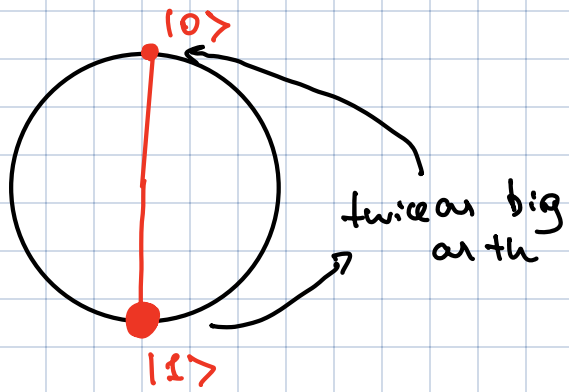
- for one qubit:
  - the "north pole" represents state $|0\rangle$,
  - the "south pole" represents state $|1\rangle$
  - the size of the blob is proportional to the prob. of measuring the respective state.
  - the colors indicate the relative phases compared to state $|0\rangle$.





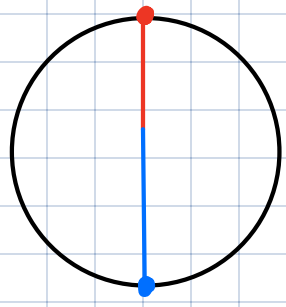
Consider:

$$\frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$$



Consider: $(|0\rangle - |1\rangle) \cdot \frac{1}{\sqrt{2}}$

$e^{i\varphi} = -1$



We can generalize this for n-qubits, there are 2^n basis state.

for $n=3$ we have:

000		
001	010	100
011	101	110
111		

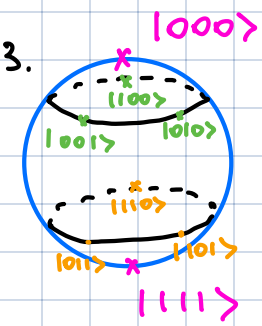
$\Rightarrow$ we plot these basis states as equally distributed points on a sphere with $0^{\otimes n}$ on the "north pole"

$1^{\otimes n}$ on the "south pole"

and all the other states are aligned on parallels, s.t the number of "1"s on each latitude is constant and increasing from North to South.

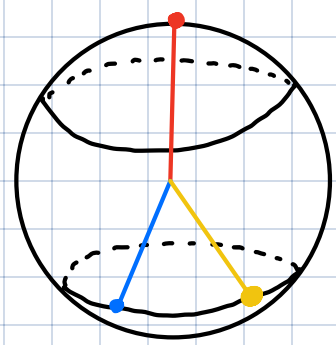
Example.

for $n=3$.



$\Rightarrow$ the size & the color of the blobs as before.

e.g $\frac{1}{2}\left(|000\rangle - |011\rangle + \sqrt{2}\cdot i \cdot |101\rangle\right)$



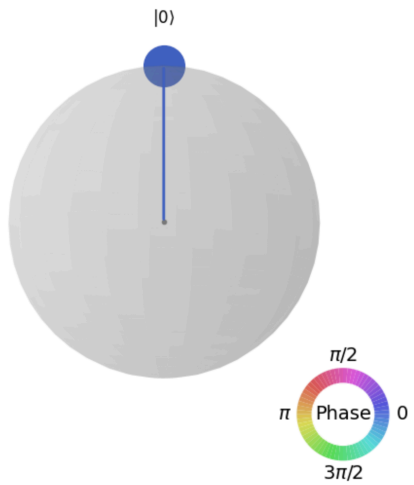
Basically the Python Code automates that.

1. Created a new Conda Environment
2. Initialized a git repo
3. Installed the necessary pip dependencies.

↳ the Result is something like this:

[1]:
```python
import numpy as np
from qiskit import QuantumCircuit
from qiskit.quantum_info import Statevector
from qiskit.visualization import plot_state_qsphere
```

[2]:
```python
n = 1
qc = QuantumCircuit(n)
statevec = Statevector.from_instruction(qc).data
print(statevec)
plot_state_qsphere(statevec)
```

```
[1.+0.j 0.+0.j]
```

[2]: